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PABEL: Plugin ABE for LLM

Ensuring Document Confidentiality for AI Agents via Attribute-Based Encryption

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1 Introduction

1.1 Context and Motivations

The constant and rapid evolution in the field of Artificial Intelligence has result in a day-to-day use of AI agents, thus the need of security and privacy when dealing with tools that, potentially, could harvest personal or company data for a variety of reasons. More in detail the concept that we were trying to focus on, was the scenario in which an employee, by chance or intentionally, had its AI agent to look at a confidential file which the company doesn't want to be leaked or simply does not trust the AI agent's vendor.

This particular fear, to not trust AI vendors, it's not unfounded; several recent incidents show how sensible data can end up outside the boundaries an organization intended to enforce, or how AI providers themselves may proceed to process user data in ways that raise legitimate concerns.

A well known example is the 2023 Samsung incident, where some employees pasted proprietary source code into a public LLM chatbot for debugging purposes, effectively transmitting confidential IP (intellectual property) to an external, unintended infrastructure. This incident led Samsung to ban all AI tools altogether[1]

A second, more institutional, example involves OpenAI: in december 2024, the Italian Data Protection Authority fined OpenAI for €15 million, finding that ChatGPT was processing users' personal data to train underlying models without the legal basis required by the GDPR; not to mention a security breach, that took place in march 2023, which exposed users' data and payment information[2].

Also Anthropic was acting in regard of the guarantees that have to be assumed by a vendor, in particular it had to update its customer terms[3]. They explicitly stated that each conversation made with Claude could be used for model training, with the exception of Claude for Work, Claude Enterprise, Claude Gov and any API-based access[3].

These three exapmles share a common thread, that is once data leaves an organization's boundary and reaches a third-party model or vendor, the organization has no further technical control over it. This thesis takes the position that this control should not depend on trusting the AI agent's behavior, nor on trusting the vendor's policies, but should instead enforced cryptographically, when accessing the data.

1.2 Activities done and Results

In this thesis, we have designed and implemented a plugin for AI agents that allows to decrypt files using Attribute-Based Encryption (ABE) in order to ensure that only authorized users and authorized AI Agents can access specific parts of the file. The plugin is designed to be easily integrated into existing AI agents, providing a seamless way to enhance security and privacy in AI-driven environments.

Such security enhancement is achieved by encrypting files with ABE, which allows for fine-grained access control based on user and AI agent attributes. PABEL intercepts file access requests from the AI agent, authenticates the user and checks, via server, if both the user's attributes and the agent's attributes match the encryption policy; thus the server decrypts the specific parts of the file that both the user and *that* agent are authorized to see and returns it to the agent.

This approach ensures that sensitive information remains protected when accessed by AI agents, since every sensitive information is handled server-side and the agent does not perform any sort of decryption and does not even hold any cryptographic material needed to do so.

The result is a working end-to-end system: a CP-ABE capable MCP server, an agnostic connector that has been verified against multiple agents and a hardened workflow to ensure agents' identites.

1.3 Technologies and Tools

To achieve the goals of this thesis, several technologies and tools have been utilized. The core of the security mechanism is based on Attribute-Based Encryption (ABE), more in particular we have used the open source library **OpenABE**[4], which we had to upgrade to a more recent version by updating its dependecies and fixing some bugs that were present in the original code.

Another important component of the system is the authentication and authorization mechanism, which is implemented using **Keycloak**, an open-source identity and access management solution. Keycloak provides a robust framework for managing user identities, roles, and permissions, which is essential for enforcing access control policies in the context of ABE.

The plugin is designed to be compatible with AI agents that support plugin integration,

and it has been tested with specific AI agents and harnesses to ensure proper functionality. The AI agents used for testing are:

- **Claude Code**, an AI agent developed by Anthropic;
- **Codex**, an AI agent developed by OpenAI;
- **Copilot for VSCode**, an AI agent developed by Microsoft, integrated into the Visual Studio Code IDE.
- **OpenCode**, an open-source AI agent.

Throughout the project we used Python as main language, as well as Python libreries such as **PyJWT**, **FastMCP** and **Psycopg**. The server-side component of the system was containerized using **Rancher Desktop**, a tool that allows to run Kubernetes clusters on a local machine. This setup enables easy deployment and management of the server component, ensuring scalability and reliability in handling decryption requests from AI agents. Alongside the server container, there were deployed two other containers: one for the Keycloak, which contains identities and attributes, server and another for a **PostgreSQL** database, which (in combination with Keycloak) store agent installation's information and generated keys. This containerized approach simplifies the deployment process and allows for an easy scaling of the system as needed.

1.4 Thesis outline

After a description of the motivation and also a security overlay of this work in section 2, we'll cover some background informations in section 3, especially Attribute-Based Encryption, before heading in a in-depth analysis of the System Design in section 4. Will follow a Security Analysis in Section 5, alongside some evaluations and the conclusions, respectively, in Sections 6 and 7.

2 Motivation and Threat Model

2.1 Problem Statement

The main problem that this thesis aims to solve comes from a consideration while thinking how AI can be implemented in a blue team scenario. Even if this topic could be an argument of discussion by itself, many perplexities arise when we think how data is exchanged considering AI agents actively inside the attack surface.

In particular, the problem is that AI agents are not able to distinguish between sensitive and non-sensitive data and, furthermore, they are not able to understand the context of the data they are processing. This means that if an AI agent is given access to sensitive data, it could potentially leak that data to an attacker, either intentionally or unintentionally. Of course, it is imperative to also consider the intrinsic vulnerability of AI agents to prompt injection attacks [5], which could be just another mean to the exfiltration of sensitive data as aforementioned.

One method to mitigate this problem would be to utilize cryptography, but that alone comes with its own set of challenges. More in detail, both an AI agent and a user cannot benefit from data that is encrypted with a symmetric key: both of them would require a decryption of said data, otherwise they would not be able even to understand it, let alone use it.

One possible alternative could be Homomorphic Encryption and, surely, it is a very promising topic, but as of today its computational limitations make it not suitable for a real-world scenario, especially if we consider the amount of data that an AI agent should process.

Another approach to achieve more security in a scenario that involves a LLM could be Trusted Execution Environments, but here lies the need (thus the problem) to ensure that the AI agent is running in a trusted environment, which is not always possible, especially if we consider that the AI agent could be running on a remote server or in a cloud environment.

Hereby we'll expand on the limitations of both Homomorphic Encryption and Trusted Execution Environments, to better understand and define the problem we are trying to solve.

2.1.1 Limitations of Homomorphic Encryption

As aforementioned, Homomorphic Encryption (HE) is a very promising topic, given the ability it enables to perform homomorphic operations on encrypted data. The problems arise when we approach statistical computations, which is the basis of Large Language Models, and, in general, of any AI agent.

Precisely on this regard, Cheon et al. eloquently stated [6]:

> Most of real-world data contain some errors from their true values. [...] an approximate value may substitute the original data and a small rounding error does not have too much effect on computation result [...] Unfortunately this rounding operation has been considered difficult to perform on HE since it is not simply represented as a small-degree polynomial.

They proposed a scheme that allows to perform approximate computations on encrypted data, not without open problems, like the need to convert said scheme in a fully homomorphic scheme using bootstrapping. This problem was later addressed by De Castro et al. who proposed a GPU-based workflow by extending the capabilities of OpenFHE [8].

Said implementations is undoubtly a great step forward towards practical encrypted LLM, but it has still non trivial limitations:

> Although promising given the massive overheads involved in both LLMs and FHE (Fully Homomorphic Encryption), this is still far from practical for real-world usage, even for applications that are not latency sensitive such as text summarization or content generation (in contrast to chatbots or Q/A tasks, which are more demanding in terms of responsiveness).

The main problem is that, even with a GPU-accelerated implementation, the bootstrap overhead (also in combination with the LLM inference overhead) is still too high to be considered practical in a every day scenario. To this we have to take in account also that by approximating the functions needed to have LLM inference we are introducing a source of error that translates into a loss of accuracy in the final result. Both of this factors alone make Homomorphic Encryption, as per today, not viable to be used in application such as chatbot or coding agents and it is clear why Attribute-Based Encryption could fill the gap in this point and time by providing security precisely where the FHE, at the moment, cannot provide.

2.1.2 Limitations of Trusted Execution Environments

Trusted Execution Environmets, as the name suggests, are trusted environment where it is possible to execute critical operation and, in the context of prompt injection and LLM, potentially separate a network or an entire device from the execution of malicious actions by the agent that, in fact, would only be relegated in a sandbox.

Geppert et al. define TEEs as following:

> TEEs [...] are defined as tamper-resistant processing environments that run on a separation kernel. Remote verification makes the technology suitable for cloud environments. It is a mechanism proving to an organization that uses the services of a cloud provider that the specified software runs on the predefined hardware even if the organization does not have direct access to this hardware.

In the cloud computing scenario this sort of infrastructure is already offered by providers like Google, Amazon, and Microsoft [9], but this would not apply to some organization that don't want to rely on external provider. This is extremely crucial even when talking of LLMs, since if an organization opted to rely on third party agents (i.e. Claude Code, Codex, etc...) it would rely to this third party also all the security assumptions that arise.

To better explain this concept, by considering the use of a TEE provided by some provider, an organization would not have any control on *how* its own data are manipulated. It is obvious that in most cases this is not an acceptable risk and many would resort to a "home-made" solution, like training its own LLM; this issue, although, could be mitigated if we consider a situation where only pre-approved data can be "sent out", thus the application of Attribute-Based Encryption and the not suitability of TEE in every scenario.

Summarizing, even though TEEs "minimizes potential security issues such as data leakage, unauthorized access, code tampering, and software-based attacks" ([10]), they do not take in consideration the cases in which a company does want to fine grain the data that escape the company's domain.

2.2 Actors and Goals

In this section we define the actors involved in the system and their goals, to better understand the problem we are trying to solve.

The actors, with their own goals, involved in the system are:

- **User:** the user is the person that wants to use an AI agent to process a document. The user has a set of attributes that define their access rights to the document.
- **AI Agent:** the AI agent is the entity that retrieves and processes the document. The AI agent has a set of attributes that define its access rights to the document.
- **Company:** the company is the entity that owns the document and defines the access policy for it, alongside the attributes of the users and the AI agents. The company is responsible for ensuring that the access policy is enforced and that the sensitive data is protected.
- **IT/Cybersec:** the company's departments responsable for the security of the documents (i.e. encrypting, authorizing and auditing) and the company allthoghether

2.3 Assumptions

We assume that the documents subject to protection are already produced and encrypted via Attribute-Based Encryption before being made available to the system described in this work. Authoring a new document, defining its access policy, ensuring the freshness of employees' attributes, generating the corresponding keys, encrypting the document, are all separate concerns and fall out of the scope of this thesis, which focuses on enforcing, at read time, a mechanism that allows an AI agent to process the document following a safe workflow, minimizing the risk of sensitive data exfiltration.

We also assume that our garantees end at the moment decrypted content is legitimately handed to the requesting AI agent, in accordance with the CP-ABE policy and the combined attributes of the user and the agent. What the agent does with that content afterwards is not something an access-control layer, operating at application level, can control and it is, therefore, out of scope. There is an exception worth to mention, that is locally materialized copies which are time-bounded and removed automatically at the end of a session, to limit possible unbounded exposure.

By design, the system assumes that the encrypted documents are `.abe` files, that is they follow a predefined structure (as defined in Section 4.2.1) that permits the recognition of encrypted files without the need to decrypt them.

2.4 Adversary Model

In this section we'll describe what entities are considered as adversary. By default, we assume the worst case scenario: both AI agent and its Vendor are not trusted, nor the user itself. Mainly, we can differentiate three different threat entities:

- **AI agent / Vendor:** the agent itself can be "too curious", it can be compromized or worse, its vendor can actively try to use customer data for its own goals (see 1.1). It's important to assume the worst case scenario, and to don't trust the model.
- **User:** both by error and intentionally, a user could try to use its AI agent to read something outside the user's authorization level.
- **Attacker:** an external entity that could also intercept traffic data and, possibly, forge malicious content.

It is imperative to understand and prevent possible data leaks, hence our proposal to put at risk only a certain part of confidential data (choosen by the company or who possess them).

It's also important to note that there are goals outside this work's scope such as:

- **OS/Endpoint already compromised:** if an attacker has already access to the machine, it is quite easy to see how this system cannot ensure its premises.
- **Secure storage of CP-ABE master key:** the CP-ABE master key is assumed to be protected by the company through appropriate measures to store it safely; its compromise would void every guarantee of the system.
- **Agent behaviour post reception of confidential data:** it is impossible for this system to control what an agent will do to a file once recieved. After the reception of the file, adequatly decrypted, it is responsability of the user what will happen to the data.

- **DoS attacks:** this system does not guarantee the availability, but the adequate confidenciality of data.

2.5 Security Goals

The end goal of this work is to provide a simple way to handle sensitive data with a fine-grained access control enforced by a company, such that its own employee could not provide leaks whenever they use an AI agent, since the access control mentioned above. This allow also to audit each and every operations on encrypted files, that is offering an invaluable tool for Cybersec/IT departments, which could easily track any eventual misbehaviour.

Our work certainly benefit corporate environments, but also any enviornment where it is wanted to limit what agents can see and therefore where more controlled access is needed, whit a consequent help on data leak control.

3 Background

3.1 Attribute-Based Encryption

Attribute-Based Encryption, as the name suggest, is a type of encryption that enforce an access control via attributes and authorization policies. The secret key of a user and the ciphertext are both dependent by attributes.

First proposed as a concept in 1984, it was improved and implemented in the first years of 2000s; it presents two main types of encription schemes: the **key-policy attribute-based encryption (KP-ABE)** [11] and the **ciphertext-policy attribute-based encryption (CP-ABE)** [12].

3.1.1 From Identity-Based to Attribute-Based Encryption

Identity-Based Encryption is a type of public key cryptography based on PKI (Public-Key Infrastructure), with the main difference being the fact that IBE allows a 1 to 1 relationship between sender and reciever. As Shamir said in his paper [13]:

> The scheme is based on a public key cryptosystem with an extra twist: Instead of generating a random pair of public/secret keys and publishing one of these keys, the user chooses his name and network address as his public key. Any combination of name, social security number, street address, office number or telephone number can be used (depending on the context) provided that it uniquely identifies the user in a way he cannot later deny, and that it is readily available to the other party.

The signature scheme abide the following verification condition [13]:

$$s^e = i \cdot t^{f(t,m)} \mod n \tag{1}$$

Where

- m is the message,
- s,t is the signature,
- i is the user's identity,
- n is the product of large primes,

- e is a large prime reltively prime to $\phi(n)$,
- f is a one way function.

IBE applications are quite immediate to understand, for example, the ability to ensure that only a given person could read a message, without relying on the PKI, is quite remarcable. There is, anyway, a trust assumtpion towards the Private Key Generator (PKG), which if compromised would nullify the secrecy of every encrypted data that use its issued keys.

Besides, IBE did point towards ABE; we can see how Sahai and Waters already introduced an ABE approach while proposing a Fuzzy Identity-Based Encryption. They did note how a system where an authority gave a user different keys for each of the attributes that describe the user could easily fall prey of collusion attacks, where multiple users combine their keys to form identities, with a combination of said attributes, that could decrypt ciphertext which none of the individual user could be able to decrypt alone [14]. This occurs by considering an identity as a set of attributes and, precisely for that, FIBE for the first time achieved error tolerance; it allows for a private key for an identity, say ω, to decrypt a ciphertext encrypted with an identity ω' if and only if $|\omega \cap \omega'| \geq d$, where d is fixed during the setup and indicates the error tolerance threshold.

Goyal et al., later proposed a better implementation that was further expanded by Bethencourt, Sahai, and Waters, de facto introducing, respectively, KP-ABE and CP-ABE.

3.1.2 Ciphertext-Policy versus Key-Policy ABE

The main difference between KP and CP ABE is their approach to the policy needed to decrypt. While in the Key-Policy ABE the key holds the policy for the identity, in the Ciphertext-Policy ABE is the ciphertext that holds said policy. In other words, for a KP approach, think of a company that encrypt all its mails with the headers as attributes, say
`{reciever: developers, object: project-x, date: 20-dec}`
When an analist, for a given reason, has to read those mails it will recieve a new key suited for the occasion, for example:
`(reciever=developers AND object=project-x) OR (date>20-dec)`
This key and this key alone can decrypt the mails that satisfy the key's rule.

Using CP ABE, instead, shift the policy on the document itself: for example, always in a corporate environment, the company encrypted a reserved memo already knowing who can read it, thus it encrypted the document with a policy like:

```
(department=finance) OR (management AND security_level > 5)
```

With this setup only those employees who are in the finance department or are in management, with a security level greater than 5, can access this document. In summa, the paradigm shift from the rule attached to the reader (KP-ABE) to the rule attached to the document (CP-ABE).

Both KP and CP, however, share the definition of a collusion resistant structure, an **Access Structure**, and four fundamental algorithms: **Setup**, **Encrypt**, **Key Generation** and **Decrypt** [11, 12]. Moreover, those implementations differ on **Encrypt**, **Key Generation** and **Decrypt**, furthermore, CP ABE includes also a **Delegate** algorithm. It is all defined as:

- **Access Structure:** Let $P_1, P_2, ..., P_n$ be a set of parties. A collection $A \subseteq 2^{P_1,P_2,...,P_n}$ is monotone $if\ \forall\ B,C\ :\ if\ B \in A\ and\ B \subseteq C\ then\ C \in A$. An **Access Structure** (monotone access structure) is a collection (monotone collection) A of non-empty subsets of $P_1, P_2, ..., P_n$, i.e. $A \subseteq 2^{P_1,P_2,...,P_n} \setminus \emptyset$. The sets in A are called the authorized sets, and the sets not in A are called the unauthorized sets.

- **Setup:** takes no input other than the implicit security parameter. It outputs the public parameters PK and a master key MK.

- **KP Encrypt:** a randomized algorithm that takes as input a message m, a set of attributes γ, and the public parameters PK. It outputs the ciphertext E.

- **KP Key Generation:** a randomized algorithm that takes as input an **Access Structure** A, the master key MK and the public parameters PK. It outputs the decryption key D.

- **KP Decryption:** takes as input the ciphertext E, encrypted under the set γ of attributes, the decryption key D for access structure A and the public parameters PK. It outputs the message $M\ if\ \gamma \in A$.

- **CP Encrypt:** takes as input the public parameters PK, a message M, and an access structure A over the universe of the attributes. The algorithm encrypt M and produce

a ciphertext CT such that only a user that possesses a set of attributes that satisfies the access structure will be able to decrypt the message. It is assumed that the ciphertext implicitly contains A.

- **CP Key Generation:** takes as input the master key MK and a set of attributes S that describe the key. It outputs a private key SK.
- **CP Decryption:** takes as input the public parameters PK, a ciphertext CT, which contains an access policy A, and a private key SK, which is a private key for a set S of attributes. If S satisfies A then the algorithm will decrypt CT and return the message M.
- **CP Delegate:** takes as input a secret key SK for some set of attributes S and a set $\tilde{S} \subset S$. It outputs a secret key SK for the set of attributes $\tilde{S}$.

In summary, we can summarize the main difference between KP-ABE and CP-ABE in a single question: *who decides who can read the document, the party that encrypts it or the party that issues the keys?* With KP-ABE the policy is embedded in the user's key, so it is fixed by the key issuer and whoever encrypts the document can only label it with attributes, with no means to control who will be able to read it. With CP-ABE, instead, the policy is embedded in the ciphertext, hence the decision belongs to whoever encrypts the document.

Given that in PABEL the document owner must define who can read the document at encryption time, CP-ABE is the natural choice and, in fact, it is the scheme adopted in the rest of this work. Moreover, even if both approaches are collusion resistant (i.e. a user with attribute "finance" and another one with "management" cannot decrypt together something that they could not read alone), the well-known limits that CP-ABE implies are a good fit for our system and actually turn out to be consistent with our setting.

First, the policy is fixed when the document is encrypted. This is acceptable here because a given policy expresses a decision taken beforehand by the company: changing it requires re-encrypting the document, an operation that neither a user nor an agent can perform. Since the decryption is not performed client-side, it also shifts the computational overhead entirely to a server.

Second, the policies being monotone, which means *access can only be granted and never subtracted by a negation*, is a deny-by-default behaviour that works perfectly in a

scenario in which we want to *specify who is permitted to read specific information*. Basically, anything that falls out of the scope prior defined by the document's owner is denied. This follows tha same reasoning behind a whitelist firewall, which is generally considered safer than a blacklist one: a blacklist *must* anticipate every possible threat, so whatever is not foreseen is allowed, whereas a whitelist enumerates only what is legitimate, hence whatever is not foreseen is blocked.

Third, revoking keys is notoriously hard, but in our system neither the user nor the agent ever holds one: keys are kept server-side, so revoking an agent is litterally a matter of deleting a database entry.

Finally, CP-ABE relies on a single central authority that holds the master key; for a company that needs a single point of control and auditing, this is a feature rather than a drawback, although it places the server inside the trusted computing base.

3.2 Identity and Access Management

Identity and Access Management (IAM) is a set of processes and technologies that establish *who* is making a request (authentication) and *what* that party is allowed to do (authorization), keeping both decisions in a single, auditible place. A modern IAM typically provides a single sign-on (SSO), multi-factor authentication and a central directory of users together with their attributes (e.g. department, role, etc...).

This last point is the one that matters the most for this work. In CP-ABE the capabilities of a user are entirely described by its attributes, so whoever is the authoritative source of those attributes is, de facto, the one deciding which key can exists. Delegating this role to an IAM means that attributes are not self-declared by the user or by the agent, but asserted by a trusted party and carried in a token. The same idea underlies attribute-based access control in general, where the authorization to perform an operation is determined by evaluating the attributes of the subject against a policy [15].

3.2.1 OAuth 2.0 and PKCE

OAuth 2.0 is an authorization framework that allows a client application to obtain limited access to a resource on behalf of a user, without ever handling the user's credentials: it is used, instead, an *access token*, which denotes a specific scope and lifetime [16]. The user authenticates against an authorization server, which issues the token to the client; the

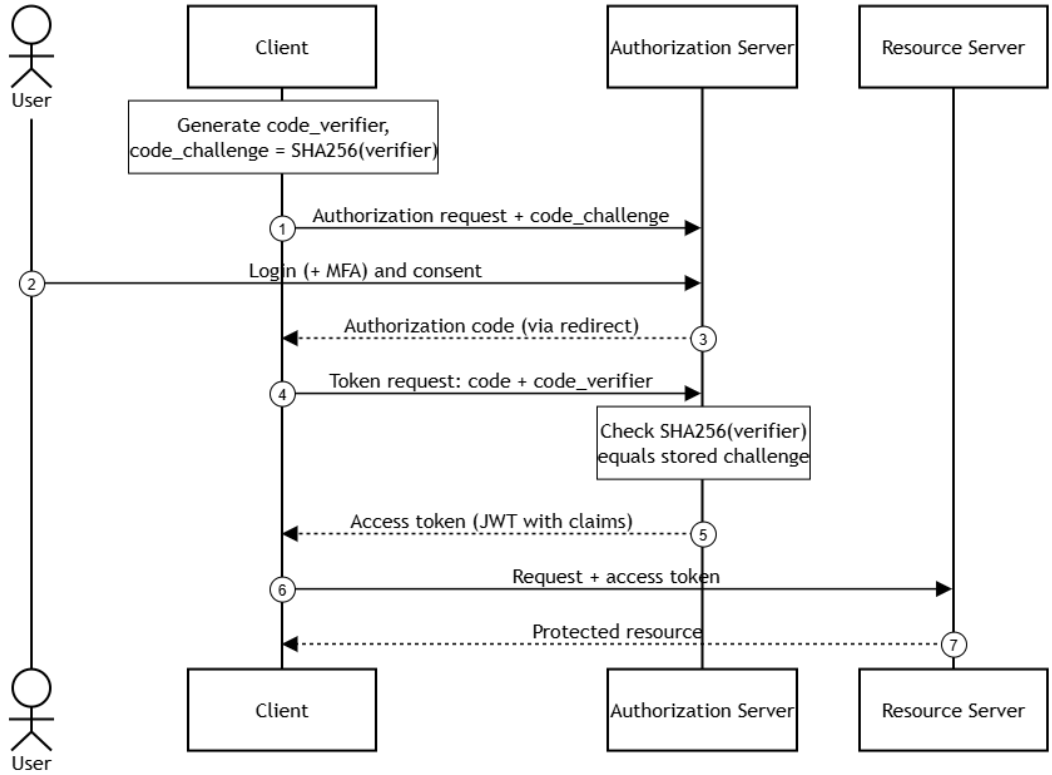


Figure 1: A typical IAM flow

token is then presented to the server that exposes the protected resource. Tokens are often encoded as JSON Web Tokens (JWT), a compact format for representing claims between parties that can be digitally signed [17] Is it through such claims that user attributes can travel from the IAM to the other parts of the system.

In the Authorization Code flow the client first receives a temporary *authorization code* through a redirect and then exchanges it for a token. For public clients, such as command line tools or mobile and desktop applications, this code can be intercepted by a malicious application on the same device and the client has no safe place to store a secret that would prevent its use [18]. Proof Key for Code Exchange (PKCE) closes this gap by generating a random *code verifier* for each client's request and sends only its trasformation

$$\texttt{code_challenge} = \text{BASE64_URL}\big(\text{SHA256}(\text{ASCII}(\texttt{code_verifier}))\big) \quad (2)$$

with the authorization request, then discloses the verifier itself only when redeeming the code [18]. An attacker who steals the code will not know the verifier and, since the use of SHA-256 as a one-way function, cannot derive it from the challenge.

Although PKCE was originally designed for native applications, is current best practise to use for all kinds of OAuth clients, including confidential ones [19].

3.2.2 Keycloak

Keycloak is an open-source IAM based on standard protocols. It supports OpenID Connect, OAuth 2.0 and SAML. At its core there are these concepts:

- *realm:* manages a set of user, credentials, roles and groups;
- *client:* an entity that can request to authenticate a user;
- *protocol mapper:* it allows to select which claims are stored in the issued tokens.

Keycloak does also support one-time password (OTP), such as Time-based OTP (TOTP) and HMAC-based OTP (HOTP) through FreeOTP. We chose it because of its open-source nature, to being consistent with the assumption of an organization who does not want to rely on third parties.

3.3 The Model Context Protocol

"Model Context Protocol (MCP) is an open protocol that enables seamless integration between LLM applications and external data sources and tools." [20]

It uses JSON-RPC 2.0 messages to establish communication between:

- ***Hosts:*** LLM applications that initiate the connection
- ***Clients:*** Connectors within the host
- ***Servers:*** Services that provide context and capabilities

More specifically, servers can offer to clients

- ***Resources:*** Context and data, for the user or the AI model to use
- ***Prompts:*** Templated messages and workflows for users
- ***Tools:*** Functions for the AI model to execute

And clients may offer, on the other end, ***Elicitation***, that is server-initiated requests for additional information from users.

However, in any situation, since the powerful capabilities through arbitrary data access and code execution, an implementor *must* carefully address **user explicit consent and** persistent **control** over what data is shared; **data privacy**, taking the appropriate measures, and ensuring that hosts must obtain explicit user consent before exposing data to servers; last, but not least, **tool safety**, since every tool represent an ***arbitrary code execution***.

For our purposes, MCP provides not only a resource server, but also the legitimate channel through which the agent asks for a document: the server owns the decryption logic and returns only the content that the policy allows. The idea is to offer to agents a black box to interact with, so they would never handle sensitive informations or at least they would not be capable to retrieve what it is not them permitted.

3.4 Hook-Based Interception in AI Coding Agents

Many coding agents expose ***hooks***, i.e. user-defined command that the agent's runtime execute at precise points of its lifecycle, such as before a tool call (`PreToolUse`), which can also block the call [21, 22]. In Claude Code, for instance, the hook receives the description of the event as JSON on standard input (tool name and arguments) and answers with a decision: allow the call, deny it with a reason, rewrite its input (`updatedInput`) or add context for the model (`additionalContext`).

The crucial property is where the hook runs. An instruction written in a prompt or in a `skill` is only text that the model reads: it may be ignored, misinterpreted or even overridden by a prompt injection [5], so it cannot be a security control. A hook is, instead, executed at runtime *outside* the model's reasoning, hence its decision does not depend on the model's cooperation. This is why **we implement enforcement as a hook** and use **skills**, informationally, **only to explain to the agent *why* a read was replaced**.

Hooks are however not uniform across vendors. Copilot CLI, for example, documents a similar `preToolUse` event with a `permissionDecision` field, but it names the field that rewrites the arguments differently (`modifiedArgs`) and supports both camelCase and PascalCase formats [22]. Moreover, some channels are not reliably delivered in practise (see the adapters in Section 4). This heterogenity motivtes the per-agent adapters

of our design.

4 System Design

4.1 Architecture Overview

Our system is composed by two main component:

- **server-side components:** the components include Identity and Access Management (**Keycloak**), database (**Postgres**), MCP server that handles logic (with **OpenABE**).
- **client-side connectors:** adapter and hook grant connection and control over the AI agent.

The server-side components are designed to be each one indipendent from the others; we have containerized each one of them with `containerd` and made them available with Rancher Desktop.

Client-side there are two parts essential to the correct flow of the program: there is an adapter for each agent and a hook that is the real intermediary between the server-side and the agent.

4.2 Early Prototyping: File Viewer and .abe format

4.2.1 File Structure

To start approaching the concept, it is imperative to subdivide the problem into smaller ones. The first problem that came to mind was to find a way to have into a single file multiple parts, each with its own encryption. To solve this problem, we did create a JSON structure that could contain each encrypted section of the file, i.e. an array containing a single object per section that has two properties: the policy requested to decipher the ciphertext and the ciphertext itself.

Listing 1: .abe format v1

```
{
  "format": "abe-doc",
  "version": 1,
  "sections": [
    {
      "policy": "livello = x",
```

```
      "chiphertext": CIPHERTEXT_Lx
    },
    {
      "policy": "livello = y",
      "chiphertext": CIPHERTEXT_Ly
    },
    {
      "policy": "livello = z",
      "chiphertext": CIPHERTEXT_Lz
    }
  ]
}
```

It's quite easy to point out that this implementation is very inefficient when the number of sections grows. To optimize the amount of operations needed and the structure of the document itself, it was improved in the version 2 of the structure:

Listing 2: .abe v2

```
{
  "format": "abe-doc",
  "version": 1,
  "sections": [
    {
      "group": 0
    },
    {
      "group": 1
    },
    {
      "group": 2
    },
    {
      "group": 1
    },
    {
      "group": 0
    }
  ],
  "groups": [
```

```
    {
      "policy": "livello >= 3",
      "ciphertext": CIPHERTEXT_L3
    },
    {
      "policy": "livello >= 2",
      "ciphertext": CIPHERTEXT_L2
    },
    {
      "policy": "livello >= 1",
      "ciphertext": CIPHERTEXT_L1
    }
  ]
}
```

This structure provides one key feature: the ability to call the decrypt function n times, where n is the number of unique attributes used to rules the sections. This is a more sustainable approach than v1, since, in that case, the decrypt function would be called exactly m times, where m is the amount of each section in the document. It's a fair assumption to say that m, *usually*, is greater than n.

4.2.2 File Viewer

Having a structured document by itself was not sufficient for our testing needs and, having the need to read and edit a `.abe` file, it was necessary to create such application. The file viewer operates as a simple file editor, with the ability to encrypt and decrypt a file using a user's private key (N.B. this feature would negate entirely the Keycloak IAM workflow and was created only for test). This private key, as seen before, has to be created and stored beforehand. A user could also authenticate itself via **Keycloak** login and so a token with its attribute will be elaborated and used in substitution of the user's key. The user-intended workflow is to login with **Keycloak** and only then he'd have access to the decryption of sections to him destinated.

There is an implicit security system, that is any user with the "security level" (an attribute defined to test a hierarchy) less than the maximum security level used in the file cannot overwrite the file.

4.3 CP-ABE Authority and Key Management

It is clear that we have to have a central autorithy that could audit and grant accesses to documents. This operation is made by conjuncting both **Keycloak** and **Postgres**: While **Keycloak** holds every user's informations (e.g. mail, department, security level, etc...), and so its attributes, **Postgres** holds a list of allowed agents with each own key.

The concept is simple: first a user get its token from Keycloak, then a cached and hashed key is kept in memory in a trusted sql database, and then, if in said database is found a key corresponding to the user's agent, the request is accepted by the server who will run the **OpenABE** operations to serve the decrypted file. With this solution nobody needs to have installed its own key and it is strictly handled "in background", to both minimizing the risk of tampering and also to ensure a strict control over the allowed agents (it's easy as deleting a row in a db to disable entirely an agent from being used).

4.4 Human Identity via Keycloak

Obviously, the Keycloak login can and has to be used with two-factor-authentication (it can also enforce a biometric login). The 2FA itself is out of this scope, but we mention **FreeOTP** as a free tool used for an otp testing.

Once the user is logged in, and we can assume it is authenticated, then we can proceed to concede a token, containing the relevant attributes, that will be used to ask for the key of said user.

4.5 Agent Identity Model

In the adversary model of this work the AI agnet is not trusted, so a policy that mentions the agent cannot rely on anything the agent declares about itself. The identity of the agent *must* be established the same way as the identity of the human, that is by a credential that the agent cannot forge, verified by the server on every call.

4.5.1 What is Identified: Harness, Surface and Model

Three different things are usually called "the agent":

- **Harness:** the product that owns the tool-call loop and decides whether a file read happens (e.g. Claude Code, OpenCode)

- **Surface:** it is where the harness runs (a terminal, an IDE extension, a desktop application) and only changes where configuration files live
- **Model:** it is what reasons inside the harness (e.g. Sonnet, GPT, etc...)

PABEL identifies the harness and nothing else: it never sees the model, and no policy can name it. A clause naming a given harness is satisfied by an installation of that harness regardless of which model is running in it, and is not satisfied by a different harness driving the same model. The unit of trust is therefore the installed product plus the person using it.

It is important to notice how the enumeration of the model and the surface of an agent is a challenge by itself. It is also important to remind that PABEL operates at layer 7, thus the impossibility to have an exact picture of what is running on the system; this could be theoretically achieved by implementing an enforcement at OS level.

4.5.2 Per-Installation Credentials

Each installation of a harness on a user's machine is registered by an administrator and receives its own Keycloak client, which needs to be authenticated with the OAuth 2.0 `client_credentials` grant. The `client_secret` is shown once at creation and handed over to the employee out of band. The credential is never reused across two harnesses: sharing it would give them a shared identity, which is exactly what this model prevents.

On every call the server verifies the agent token as it does for the human one (signature, issuer and expiration) and then resolves the identity through a chain that starts from a value that the caller cannot choose:

1. the `azp` (authorized party) claim of the token, i.e. `client_id` that obtained it, stamped by Keycloak at issuance;
2. the matching `agent_installations` row, confirming that the installation exists and has not been revoked;
3. the row of the `agents` table of the product it belongs to, which tells wheter the product is enabled, which attribute it contribuites (e.g. `agent_claude_code`, `agent_opencode`) and which role a user must hold to use it.

The `agent_id` comes out of this chain only after the token has been verified: it is never a parameter of a request, so no caller can claim one.

Keycloak roles are very important and useful to handle specifically *who* can use *what*; someone without the role `agent_claude_code_user` could not integrate the agent's attributes, i.e. policies that enforces `agent_claude_code` would fail.

4.5.3 Authorization of the Pair

There are two main types for a request denial when trying to authorize an agent.

- An *hard error* is considered whenever there is an invalid (or expired) token or when it is present an unknown or revoked installation and a disabled product, hence the request is simply denied.
- A *soft error* present itself if the installation is valid, but the user doesn't hold the role that the product requires (e.g. `agent_claude_code_user`); the failure, in this case, resolves in the agent that contributes with no attribute at all. The same user is still authenticated, but any policy that names the agent will fail, so no special deny logic is required: not holding the role *is* the denial.

So, whereas there is no *hard error*, the attributes of the user and the one of the agent are then merged togheter in a single CP-ABE key (see Section 4.3). A section sealed under a policy such as `sales and agent_claude_code` will open only for someone holding `sales` ***and*** working through a registered Claude Code installation; changing either half keeps the section closed.

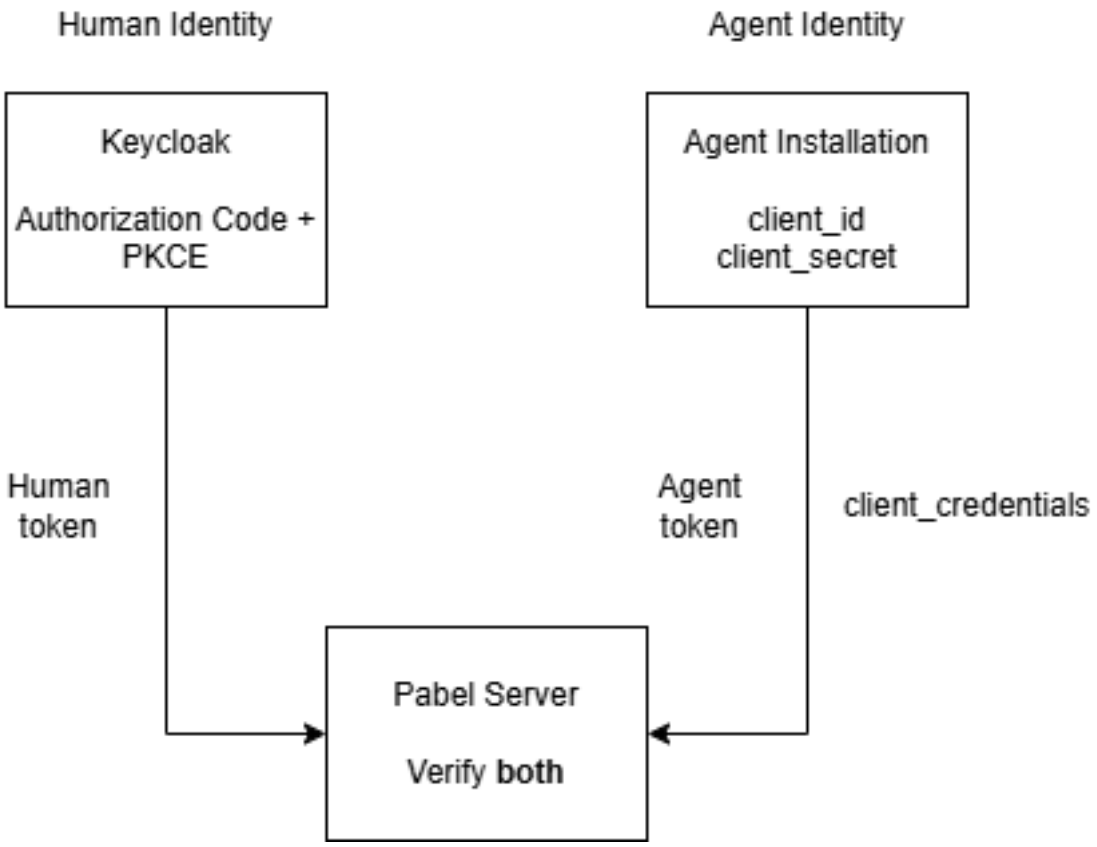


Figure 2: The two identities verified by the server on every request

4.5.4 Where the Identity Comes From and its Limits

As previously mentioned, the agent never holds its own token but it's the hook, instead, that injects the credential of the installation into the call (later expanded in The Relay Hook); the model does not need it and cannot read it. The strength of this identity is, however, bound to the protection of the credential store on the employee's machine. Whoever can read it can authenticate as that installation from anywhere; this is the compromised-endpoint scenario that is declared out of scope in the threat model.

4.6 Audit Logging

Each and every operation the agent make while calling the server is logged, both to log every access to each document and to ensure their confidentiality. Server-side it is all kept in a `audit.jsonl`, for semplicity's sake, but it can be enhanced.

4.7 Connector Architecture

The shared core evaluates every tool call against the checks of Table 1 in the order shown, and the first one that applies decides. Every outcome is also written to a local log on the client machine and the denials that would otherwise never reach the server are reported in it, so that it is kept an audit trail.

Check	**Decision**	**Why**
The call targets the local credential store	`DENY_CREDENTIAL_ACCESS`	secrets are used only by the connector's own code
A command references the connector's internals	`DENY_HOOK_BYPASS`	the hook is the only sanctioned way to run it
Direct call to the local PABEL tools	`ALLOW`	sanctioned call
Direct call to the deployed server's tools	`ALLOW`, agent token injected	the model never holds the credential
Write to a protected document	`DENY_MUTATING`	PABEL has no authoring path
Write to the hook configuration	`DENY_CONFIG_TAMPER`	it would let the installation act as another agent
Command invoking the OpenABE binaries	`DENY_OABE_BINARY`	it would bypass the combined key and the audit
No protected target is mentioned	`ALLOW`	not a PABEL concern
Ambiguous target (directory, pattern, several files)	`DENY_AMBIGUOUS`	it cannot be relayed automatically
One protected file	`DENY_WITH_RELAY`, or `DENY_AUTH_ERROR` / `DENY_RELAY_ERROR` on failure	the direct read is replaced by the server's result

Table 1: Decisions of the shared core, in evaluation order

4.7.1 The Relay Hook

4.7.2 Per-Agent Adapters

Since each agent has its own way to be controlled, it is necessary to create a specific adapter for each one of them. The adapter is the component that will translate the generic workflow into agent-specific ones. The general approach is to fold the content of the decision into the agent's reasoning process, so that if the decision is `None`, or better, if there is a deny from the `core.decide` function, then the agent will not be able to access the document and so it will not be able to use it in its reasoning process. This

is important because each and every agent has its own way to reason and use hooks, for example, Claude code uses `decision.updated_input`, thanks to which we can impose the use of our tool; different story is for Copilot CLI we cannot fully rely to `additionalContext`.

Table 2 summarises, for each harness, how a denial travels back to the agent and where the relayed content lands.

Harness	**Blocking channel**	**Content channel**
Claude Code	`permissionDecision`	`additionalContext`
OpenCode	exception thrown from `tool.execute.before`	folded into thrown message
Copilot (VS Code)	`permissionDecision`	`additionalContext`
Copilot CLI	`permissionDecision`	folded into `permissionDecisionReason` (`additionalContext` is unreliable)
Cursor	`permission`: alow, deny or ask	`agent_message`
Codex CLI, ChatGPT desktop	none (no hook)	none

Table 2: How each harness receives the denial and the relayed content

Following we can see fragments of those two implementations for Claude code and Copilot CLI, respectively:

Listing 3: Claude code adapter

```
def render(decision: Decision) -> RenderedResponse:
    if decision.kind == DecisionKind.ALLOW:
        if decision.updated_input is None:
            return RenderedResponse()  # no output at all - the tool
                call proceeds untouched
        # A direct call to pabel's own tools, allowed through with this
        # installation's agent credential injected - see core/decide.py
            .
        return RenderedResponse(stdout=json.dumps({
            "hookSpecificOutput": {
                "hookEventName": "PreToolUse",
                "permissionDecision": "allow",
                "updatedInput": decision.updated_input,
            }
```

```
        }))

    output = {
        "hookSpecificOutput": {
            "hookEventName": "PreToolUse",
            "permissionDecision": "deny",
            "permissionDecisionReason": decision.reason,
        }
    }
    if decision.content is not None:
        output["hookSpecificOutput"]["additionalContext"] = json.dumps(
            decision.content)
    return RenderedResponse(stdout=json.dumps(output))
```

Listing 4: Copilot CLI adapter

```
def render(decision: Decision) -> RenderedResponse:
    if decision.kind == DecisionKind.ALLOW:
        return RenderedResponse()

    # The guaranteed channel: fold the relay result into the reason
        text
    # itself, since additionalContext delivery is unreliable here.
    reason = fold_content_into_reason(decision)
    output = {
        "hookSpecificOutput": {
            "hookEventName": "preToolUse",
            "permissionDecision": "deny",
            "permissionDecisionReason": reason,
        }
    }
    if decision.content is not None:
        output["hookSpecificOutput"]["additionalContext"] = json.dumps(
            decision.content)
    return RenderedResponse(stdout=json.dumps(output))
```

4.7.3 Two Classes of Agents

Agents differ on what their harness lets us control and this splits them into two categories:

- **Interceptable agents**, which expose a hook that the runtime executes before a toll call; a direct read of a protected file can be blocked and replaced by the relayed, access-controlled result, without the model's intervention. Two example are Claude Code and OpenCode.

- **MCP-only agents**, i.e. agents that provide no interception mechanism. PABEL tools are registered as ordinary MCP tools that the model is free to call or not. They are accompanied by an informational skill, which is guidance and not an enforcement.

It is important to state what this distinction does *not* change, in fact, a protected file is encrypted in both cases, so its confidentiality does not depend on the hook.

What the hook really adds is the impossibility to bypass the workflow, for example, by reaching the credential store, rewriting the hook configuration, invoking the OpenABE binaries directly (if available) or even lying about the agent's own identity. For an MCP-only agent these protections are not available and the agent is simply a client for the server like any other.

The status of each harness is reported in Section 6

4.8 Document Lifecycle

The document starts its life whenever it is created and encrypted by somenoe, say a manager, that has to include sensitive information in said document. Once the appropriate policy is defined and used to encrypt the document, it is stored and it become available through requst at the MCP server.

When an agent requests access to a document, the hook will intercept the request and will start the authentication, first of the user and then of the agent. Everything is used to evaluate the CP-ABE policy and, if the user and the agent are both allowed to access the document, then the decryption will be performed and the document will be sent back to the agent; if not, the request will be denied and the agent will not have access to the document.

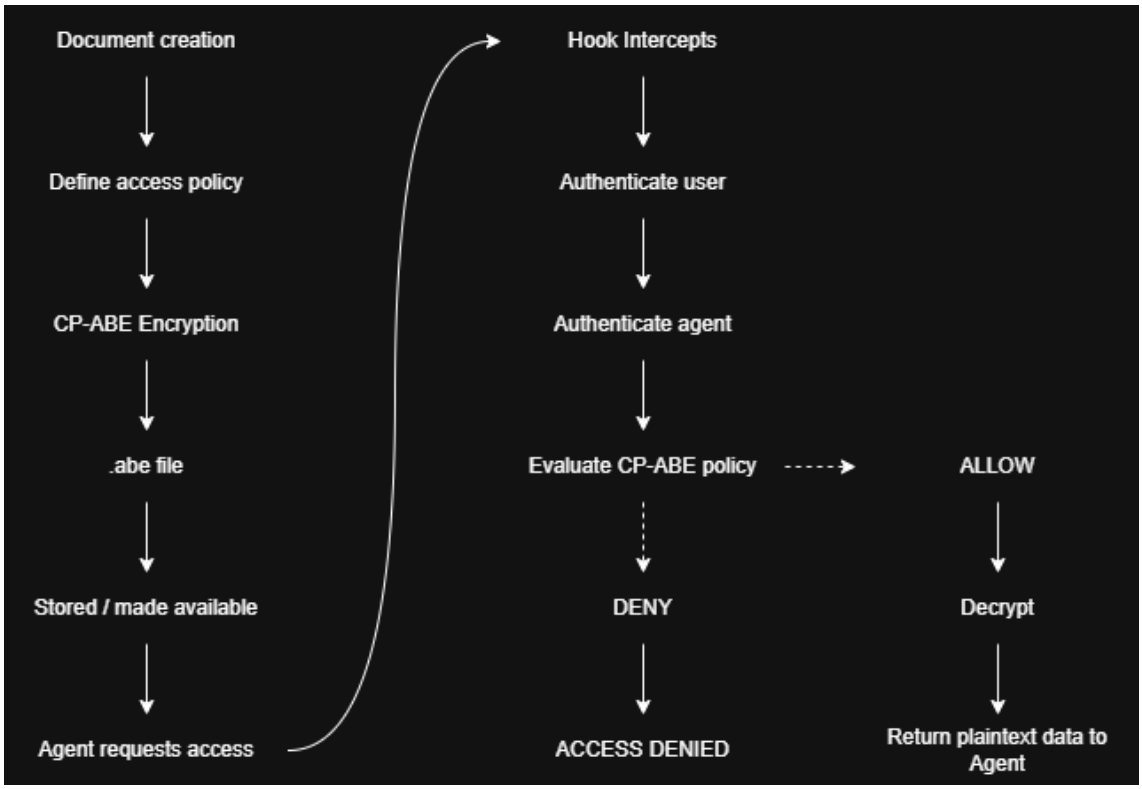


Figure 3: The flow diagram of the document lifecycle

4.9 Deployment Architecture

We hereby explain how the architecture is topologically composed and, more in detail, where each component stores its data.

4.9.1 Topology

PABEL is deployed on two kinds of machines: the **server host** and the **employee machine**. The first one runs three containers (Keycloak, PostgreSQL, MCP server) orchestrated with `nerdctl compose` on Rancher Desktop and it is managed by an admin, the second one runs the agent alongside the connector (hook, adapters and local credential store). The system was also tested with server and client on two different machines.

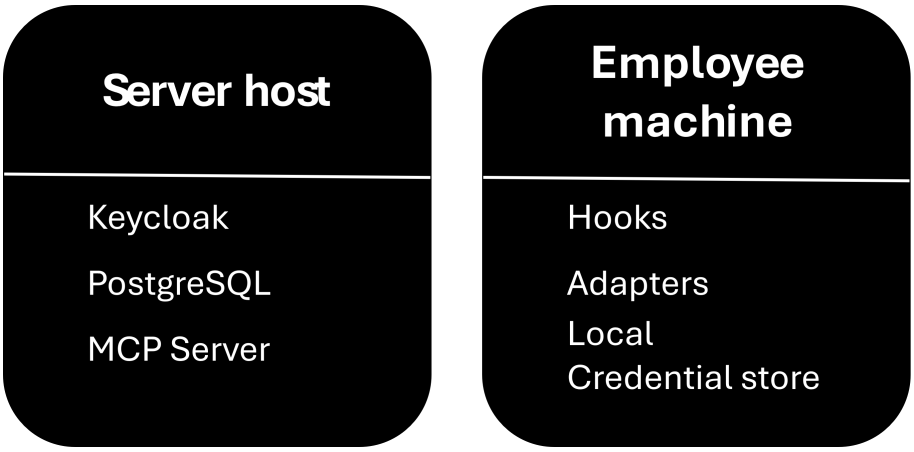


Figure 4: The two types of machines

It is worth note that only the container with the MCP server holds the CP-ABE authority and each employee machine reaches the server over the network.

4.9.2 Location of Each Component

The following table (Table 3) lists, for each component, the state it holds and where.

Component	Runs on	State	Persistence
Keycloak	server host	users, attributes, roles, agent clients	inside the container, no volume
PostgreSQL	server host	`agents`, `agent_installations`, `agent_keys`, `audit_log`	volume `pgdata`
MCP server	server host	master public and secret keys, `audit.jsonl`	bind mounts `authority` and `state`
Connector	employee machine	agent credential, human session	user directory `~/.pabel/`

Table 3: State held by each component and where it is stored

5 Security Analysis

5.1 Security Properties

Threat	***Attack Goal***	***Pabel Mitigation***
Unauthorized user	Read protected data	Human authentication + ABE
Unauthorized agent	Read protected data	Agent identity + ABE
User / Agent mismatch	Bypass policy	Combined attributes
Agent Impersonation	Fake agent identity	Installation credentials
Prompt injection	Trick agent into requesting secrets	Interception + authorization
Network attacker	Steal / forge traffic	Token verification + TLS [+ VPN]
Compromised endpoint	Read plaintext	Out of scope
Master Key compromise	Decrypt everything	Operational key protection
Post-decryption leakage	Exfiltrate plaintext	Out of scope / limitation

Table 4: Summary of various possible threat scenarios

5.2 Unauthorized User

5.3 Unauthorized Agent

5.4 User-Agent Collusion

5.5 Prompt Injection

5.6 Agent Impersonation

5.7 Credential Theft

5.8 Network Attacker

5.9 Compromised Endpoint

5.10 Master Key Compromise

5.11 Post-Decryption Leakage

5.12 Limitations

6 Evaluation

7 Conclusions

Conclusions

8 Bibliography

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